

INTELLIGENT BORDER VIDEO ANALYTICS PLATFORM (IBVAP)

Comprehensive Codebase Audit, Judge Requirement Gap Analysis, and Feature Addition Roadmap

Platform: v0.6 Baseline	Sponsor: MHA / BSF / SSB	Target: Hackathon Finale	Deployment: 100% On-Prem
Model: YOLOv8n + ByteTrack	Ledger: SHA-256 Hash Chain	Date: Sep 06, 2026	Status: Roadmap Active

[EXECUTIVE MANDATE] Transform legacy analog/IP CCTV cameras along Indian borders into an autonomous, alert-driven sensor grid without purchasing expensive smart cameras (Rs. 80K-2L/unit) or recurring cloud subscriptions. This audit establishes the engineering plan to bridge existing code gaps and maximize defense evaluation scores.

1. Executive Summary & Strategic Context

Border Out Posts (BOPs) operated by BSF, SSB, and ITBP manage thousands of legacy cameras, yet face rapid human vigilance collapse after 20-30 minutes of continuous screen watching. Infiltration, smuggling, and drone activity are discovered only post-facto. While vendors push proprietary smart cameras costing Rs. 80,000 to Rs. 2,00,000+ per unit (Rs. 8-20 Cr per 1,000 cameras), IBVAP introduces a pure edge-software gateway. By processing RTSP feeds locally on a single Rs. 2.5L edge GPU server per 10-16 camera cluster, IBVAP delivers 20-50x capex savings, eliminates foreign OEM lock-in, and enforces Atmanirbhar Bharat defense autonomy.

2. Forensic Audit of Current Codebase (v0.6 Baseline)

Module / File	Implementation	Current Strengths	Critical Gaps & Vulnerabilities
code/live.py	YOLOv8n CPU detection loop.	320x320 downsampling (~18 FPS CPU).	CRITICAL BUG: Uses cv2.CAP_DSHOW. Fails on macOS/Linux.
code/track.py	YOLO + ByteTrack tracking.	40-pt trails; px speed; track age.	No real-world metric calibration; naive 10s track purge.
code/fence.py	OpenCV click-drawn polygon fence.	Sub-ms pointPolygonTest; tripwire logic.	STRESS-TEST FLAW: Drawn on CV window; lost on restart; no Web UI.
code/rule.py	Integrated pipeline & scoring brain.	Explainable score; cooldowns; animal filter.	Single-threaded; naive night check; lacks CLAHE/MOG2 fallback.
code/ledger.py	SHA-256 tamper-evident ledger.	TOP DEFENSE USP: Chain verify + anchor head.	Needs exportable Section 65B court evidence certificate PDF.
code/dashboard.py	FastAPI dark command centre UI.	Data polling; live ledger badge; audio beeps.	4s polling latency; no live video canvas; no map; no zone drawing.

3. Requirement vs. Codebase Gap Matrix (MHA / BSF Scoring Rubric)

Capability	Evaluator Expectation	Current State (v0.6)	Severity	Required Action
Multi-RTSP Streams	8-12 streams concurrent; auto-reconnect.	Single-stream sync loop; crashes on Mac.	CRITICAL	Threaded queue stream manager + MediaMTX harness.
Dynamic Zones	Web UI polygon drawing; DB/JSON persistence.	OpenCV window click; erased on quit.	CRITICAL	HTML5 Canvas polygon editor + REST API to zones.json.
Choke-Point FRS	5-10m gate recognition; NIST pixel realism.	Completely absent in codebase.	MAJOR	YuNet/RetinaFace + MobileFaceNet gallery matching.
Check-Point ANPR	Indian vehicle plate detection & OCR.	Completely absent in codebase.	MAJOR	YOLO plate detector + PaddleOCR with regex filter.
Night Vision	IR noise filtering + low-light enhancement.	Brightness mean check only (+25 pts).	HIGH	Dual-Path: CLAHE contrast booster + MOG2 motion gating.
Anti-Fatigue Triage	<5 actionable alerts/cam/day; dedup.	Cooldowns (8s/3s) + animal suppression.	PASS	Expose suppression analytics counter on dashboard.
Evidence Ledger	SHA-256 hash chaining; court validity.	Fully working in ledger.py with verify.	PASS	Exportable Section 65B Indian Evidence Act certificate.
C2 Integration	REST/Webhook API for external VMS/C2.	Basic GET /api/state for frontend.	HIGH	Asynchronous Webhook POST + OpenAPI /docs swagger.
Tactical UI / Map	Live video stream; 2D BOP digital twin map.	Static snapshot feed with 4s poll.	HIGH	MJPEG live video canvas + 2D BOP tactical map.

4. Defense Evaluator Stress-Test Q&A; Pre-Emption

Judge Stress Question	Hidden Trap	Winning Engineering Response
Q1: 'You still need an edge GPU server—isn't that the same cost?'	Testing capex math understanding.	'No, Sir: Smart cameras cost Rs. 80k–2L per unit × 1,000 cams (Rs. 8–20 Cr). One Rs. 2.5L edge GPU server runs 12–16 streams, cutting capex by 25–50x while eliminating vendor lock-in.'
Q2: 'Infiltration happens at 2 AM in IR. Daylight YOLO fails. What happens?'	Exposing daylight-only models.	'We run a Dual-Path Night Engine: CLAHE enhances low-light edges, and an MOG2 motion-differencing safety net trips zone alarms even if neural confidence drops in pitch darkness.'
Q3: 'Faces at 200m are 15px. Is FRS science fiction?'	Detecting fake optical claims.	'FRS is strictly scoped to Choke-Point Gates (5–10m, 80–120px) under NIST FRVT standards. At perimeter distance (50–300m), we use Appearance-Based Re-ID and trajectory tracking.'
Q4: Live Challenge: 'Draw a fence on Cam 2 right now from UI without code.'	The fatal prototype killer.	'Demonstrated live in 5 seconds: Operator clicks Configure Zones on Web UI, draws polygon on canvas, clicks Save. REST API writes to zones.json and updates inference in memory.'
Q5: 'Animals trigger 200 false alarms a night. How to avoid fatigue?'	Operational reality test.	'COCO animal classes are filtered to background logs with 0 alarm. Track-ID cooldowns group loitering into one consolidated dossier. Target: <5 actionable alarms/cam/day.'

5. Architectural Blueprint: 9 Core Feature Additions

- **1. Multi-Stream RTSP Concurrency Engine:** Threaded VideoStreamManager with ring buffer queues and keyframe decimation (6 FPS analysis, 25 FPS display) supporting 8-12 streams per GPU/CPU node.
- **2. Config-Driven Dynamic Web Polygon Editor:** HTML5 Canvas polygon drawer with REST endpoints (GET/POST /api/zones) persisting to data/zones.json for live runtime boundary modifications.
- **3. BOP Gate ANPR Engine:** YOLO plate detector + PaddleOCR with Indian plate regex validation (^[A-Z]{2}[0-9]{2}[A-Z]{1,2}[0-9]{4}\$) and stolen/suspect vehicle hotlist alerts.
- **4. Choke-Point FRS (Gate Watchlist):** YuNet/RetinaFace + MobileFaceNet 512-dim embeddings for 5-10m choke-points with local encrypted SQLite gallery and DPDP Act 2023 safeguards.
- **5. Dual-Path Night Vision & IR Motion Gating:** CLAHE contrast boosting for low-light plus MOG2 motion differencing fallback inside virtual fences when neural confidence drops.
- **6. Tactical Digital-Twin 2D BOP Mini-Map:** Interactive map with camera FOV coverage cones, real-time pulsing alert beacons, and instant live MJPEG video stream popups.
- **7. Camera Tamper & Lens Occlusion Detection:** Real-time histogram shift and Laplacian variance analysis detecting camera blinding, spray paint, physical displacement, or cable cuts.
- **8. Standardized C2 Integration & Webhooks:** Outgoing HTTP POST webhook dispatcher pushing structured alert JSON to Sector HQ plus interactive Swagger documentation at /docs.
- **9. Section 65B Court Evidence Certificate:** Exportable legally defensible PDF audit dossier embedding SHA-256 chain seal, snapshot, and officer signature block under Bharatiya Sakshya Adhiniyam 2023.

6. Capex Economics: 100-Camera BOP Sector Model

Cost Dimension	Smart Camera Hardware Replacement	IBVAP Edge Software Gateway
Camera Unit Cost	Rs. 80,000 to Rs. 1,80,000 / camera	Rs. 0 (Uses existing legacy CCTV)
Total Camera Capex	Rs. 80 Lakhs to Rs. 1.80 Crores	Rs. 0 for camera replacement
Compute Infrastructure	Central NVRs / Cloud VMS: Rs. 15L–25L	10× Edge GPU Nodes: Rs. 20L–25L
Recurring Cloud Licenses	Rs. 3,000 to Rs. 8,000 / cam / year	Rs. 0 (100% On-Premise, Open IP)
TOTAL 5-YR SECTOR TCO	Rs. 1.10 Cr to Rs. 2.20 Crores	Rs. 20 Lakhs (82-91% Savings!)

7. 36-Hour Hackathon Execution Roadmap & Demo Moves

- **Hours 00 - 08 (Engine Core & Cross-Platform):** Fix macOS/Linux camera indexing; build Threaded VideoStreamManager; implement persistent data/zones.json REST API.
- **Hours 08 - 18 (Check-Point Intelligence):** Integrate PaddleOCR for Indian plates; add YuNet+MobileFaceNet choke-point FRS; implement gate alert escalation.
- **Hours 18 - 28 (Tactical UI & Night Pipeline):** Build MJPEG live video canvas; implement browser polygon drawing tool; add 2D BOP map; add CLAHE/MOG2 night path.
- **Hours 28 - 36 (C2 Webhooks & Demo Rehearsal):** Finalize C2 webhook dispatcher; test Section 65B PDF certificate; rehearse 3 killer demo moves (unplug internet, draw zone live, tamper ledger demo).

[DEMO DAY FORMULA] Walk in with this single opening sentence: *'We do not sell expensive new cameras. We turn the thousands of cameras already guarding our borders into an intelligent, alert-driven defense grid—100% offline, on-premise, tonight.'*